

Proof of designated-time stability of the controller

Step 1. Define $L_1 = (1/2)\mathbf{e}_1^T \mathbf{e}_1 + (1/2) \sum_{i=1}^n (\|\tilde{\mathbf{H}}_{i1}\|^4)$.

Case 1. $|e_{i,1}| \geq d_{i,1}$

$$\begin{aligned}
 \dot{L}_1 &= \sum_{i=1}^n \left\{ e_{i,1} \left[- \left(\eta_{i,1} + \frac{\beta_1 + \|\hat{\mathbf{H}}_{i1}\|^2}{2} \right) \Phi(e_{i,1}) - \frac{|\bar{\gamma}_{i,1}|^2 \lceil e_{i,1} \rceil}{|\bar{\gamma}_{i,1} e_{i,1}| + e_1^*} + D_{i,1} - \dot{\chi}_{i,r} \right] \right. \\
 &\quad \left. - \|\tilde{\mathbf{H}}_{i1}\|^2 \left(-\ell_{i,1} \|\hat{\mathbf{H}}_{i1}\|^2 + \frac{|e_{i,1}|}{2} \right) \right\} + \mathbf{e}_1^T \mathbf{e}_2 - \mathbf{e}_1^T \mathbf{e}_1 \\
 &\leq \sum_{i=1}^n \left\{ - \left(\eta_{i,1} - \frac{\|\tilde{\mathbf{H}}_{i1}\|^2}{2} \right) |e_{i,1}| - \frac{|\bar{\gamma}_{i,1} e_{i,1}|^2}{|\bar{\gamma}_{i,1} e_{i,1}| + e_1^*} + \ell_{i,1} \|\tilde{\mathbf{H}}_{i1}\|^4 + \frac{\ell_{i,1} \|\mathbf{H}_{i1}\|^2}{4} - \ell_{i,1} \|\tilde{\mathbf{H}}_{i1}\|^4 - \frac{\|\tilde{\mathbf{H}}_{i1}\|^2 |e_{i,1}|}{2} \right\} \\
 &\quad + \frac{\boldsymbol{\varepsilon}_1^T \boldsymbol{\varepsilon}_1}{2} + \frac{\mathbf{e}_2^T \mathbf{e}_2}{2} \\
 &\leq \sum_{i=1}^n \left\{ - \left(\eta_{i,1} - \frac{\|\tilde{\mathbf{H}}_{i1}\|^2}{2} \right) |e_{i,1}| - |\bar{\gamma}_{i,1} e_{i,1}| + e_1^* + \frac{\ell_{i,1} \|\mathbf{H}_{i1}\|^2}{4} - \frac{\|\tilde{\mathbf{H}}_{i1}\|^2 |e_{i,1}|}{2} \right\} + \frac{\boldsymbol{\varepsilon}_1^T \boldsymbol{\varepsilon}_1}{2} + \frac{\mathbf{e}_2^T \mathbf{e}_2}{2} \\
 &\leq \sum_{i=1}^n \left\{ -\eta_{i,1} |e_{i,1}| - \zeta |e_{i,1}|^{2-\kappa} \exp(|e_{i,1}|^\kappa) + e_1^* + \frac{\bar{\varepsilon}_{i,1}^2}{2} + \frac{\ell_{i,1} \|\mathbf{H}_{i1}\|^4}{4} \right\} + \frac{\mathbf{e}_2^T \mathbf{e}_2}{2}
 \end{aligned} \tag{1}$$

Case 2. $|e_{i,1}| < d_{i,1}$

$$\begin{aligned}
 \dot{L}_1 &= \sum_{i=1}^n \left\{ e_{i,1} \left[- \left(\eta_{i,1} + \frac{\beta_1 + \|\hat{\mathbf{H}}_{i1}\|^2}{2} \right) \Phi(e_{i,1}) - \frac{|\bar{\gamma}_{i,1}|^2 \lceil e_{i,1} \rceil}{|\bar{\gamma}_{i,1} e_{i,1}| + e_1^*} + D_{i,1} - \dot{\chi}_{i,r} \right] \right. \\
 &\quad \left. - \|\tilde{\mathbf{H}}_{i1}\|^2 \left(-\ell_{i,1} \|\hat{\mathbf{H}}_{i1}\|^2 + \frac{|e_{i,1}|}{2} \right) \right\} + \mathbf{e}_1^T \mathbf{e}_2 - \mathbf{e}_1^T \mathbf{e}_1 \\
 &\leq \sum_{i=1}^n \left\{ \frac{d_{i,1} (\beta_1 + \|\mathbf{H}_{i1}\|^2)}{2} - |\bar{\gamma}_{i,1} e_{i,1}| + e_1^* + \ell_{i,1} \|\tilde{\mathbf{H}}_{i1}\|^2 \|\hat{\mathbf{H}}_{i1}\|^2 - \frac{\|\tilde{\mathbf{H}}_{i1}\|^2 |e_{i,1}|}{2} \right\} + \frac{\boldsymbol{\varepsilon}_1^T \boldsymbol{\varepsilon}_1}{2} + \frac{\mathbf{e}_2^T \mathbf{e}_2}{2} \\
 &\leq \sum_{i=1}^n \left\{ \frac{d_{i,1} (\beta_1 + \|\mathbf{H}_{i1}\|^2)}{2} - \zeta |e_{i,1}|^{2-\kappa} \exp(|e_{i,1}|^\kappa) + e_1^* + \frac{\ell_{i,1} \|\tilde{\mathbf{H}}_{i1}\|^4}{2} + \frac{\ell_{i,1} \|\mathbf{H}_{i1}\|^4}{2} - \ell_{i,1} \|\tilde{\mathbf{H}}_{i1}\|^4 \right. \\
 &\quad \left. + \frac{\ell_{i,1} \|\tilde{\mathbf{H}}_{i1}\|^4}{2} + \frac{d_{i,1}^2}{2\ell_{i,1}} \right\} + \frac{\boldsymbol{\varepsilon}_1^T \boldsymbol{\varepsilon}_1}{2} + \frac{\mathbf{e}_2^T \mathbf{e}_2}{2} \\
 &\leq \sum_{i=1}^n \left\{ -\zeta |e_{i,1}|^{2-\kappa} \exp(|e_{i,1}|^\kappa) + e_1^* + \frac{\bar{\varepsilon}_{i,1}^2}{2} + \frac{d_{i,1} (\beta_1 + \|\mathbf{H}_{i1}\|^2)}{2} + \frac{\ell_{i,1} \|\mathbf{H}_{i1}\|^4}{2} + \frac{d_{i,1}^2}{2\ell_{i,1}} \right\} + \frac{\mathbf{e}_2^T \mathbf{e}_2}{2}
 \end{aligned} \tag{2}$$

Step 2. Design $L_2 = (1/2)\mathbf{e}_2^T \mathbf{e}_2 + (1/2) \sum_{i=1}^n (\|\tilde{\mathbf{H}}_{i2}\|^4)$.

Case 1. $|e_{i,2}| \geq d_{i,2}$

$$\begin{aligned}
\dot{L}_2 &= \sum_{i=1}^n \left\{ e_{i,2} \left[- \left(\eta_{i,2} + \frac{\beta_2 + \|\hat{\mathbf{H}}_{i2}\|^2}{2} \right) \Phi(e_{i,2}) + \mathbf{H}_{i2}^T \boldsymbol{\psi}_2(N_2) + \varepsilon_{i,2} \right] - \frac{|\bar{\gamma}_{i,2}|^2 e_{i,2} \lceil e_{i,2} \rceil}{|\bar{\gamma}_{i,2} e_{i,2}| + e_2^*} \right. \\
&\quad \left. - \|\tilde{\mathbf{H}}_{i2}\|^2 \left(-\ell_{i,2} \|\hat{\mathbf{H}}_{i2}\|^2 + \frac{|e_{i,2}|}{2} \right) \right\} + \mathbf{e}_2^T \mathbf{M}^{-1} \mathbf{K} \mathbf{e}_3 - \frac{3}{2} \mathbf{e}_2^T \mathbf{e}_2 \\
&\leq \sum_{i=1}^n \left\{ - \left(\eta_{i,2} - \frac{\|\tilde{\mathbf{H}}_{i2}\|^2}{2} \right) |e_{i,2}| - \frac{|\bar{\gamma}_{i,2} e_{i,2}|^2}{|\bar{\gamma}_{i,2} e_{i,2}| + e_2^*} + \ell_{i,2} \|\tilde{\mathbf{H}}_{i2}\|^4 + \frac{\ell_{i,2} \|\mathbf{H}_{i2}\|^2}{4} - \ell_{i,2} \|\tilde{\mathbf{H}}_{i2}\|^4 \right. \\
&\quad \left. - \frac{\|\tilde{\mathbf{H}}_{i2}\|^2 |e_{i,2}|}{2} \right\} + \mathbf{e}_2^T \mathbf{M}^{-1} \mathbf{K} \mathbf{e}_3 + \frac{\varepsilon_2^T \varepsilon_2}{2} - \mathbf{e}_2^T \mathbf{e}_2 \\
&\leq \sum_{i=1}^n \left\{ -\eta_{i,2} |e_{i,2}| - \zeta |e_{i,2}|^{2-\kappa} \exp(|e_{i,2}|^\kappa) + e_2^* + \frac{\bar{\varepsilon}_{i,2}^2}{2} + \frac{\ell_{i,2} \|\mathbf{H}_{i2}\|^4}{4} \right\} + \frac{\bar{k}^2 \mathbf{e}_3^T \mathbf{e}_3}{2} - \frac{\mathbf{e}_2^T \mathbf{e}_2}{2}
\end{aligned} \tag{3}$$

$\bar{k}^2$ is defined as greater than $\|\mathbf{M}^{-1} \mathbf{K}\|^2$.

Case 2. $|e_{i,2}| < d_{i,2}$

$$\begin{aligned}
\dot{L}_2 &= \sum_{i=1}^n \left\{ e_{i,2} \left[- \left(\eta_{i,2} + \frac{\beta_2 + \|\hat{\mathbf{H}}_{i2}\|^2}{2} \right) \Phi(e_{i,2}) + \mathbf{H}_{i2}^T \boldsymbol{\psi}_2(N_2) + \varepsilon_{i,2} \right] - \frac{|\bar{\gamma}_{i,2}|^2 e_{i,2} \lceil e_{i,2} \rceil}{|\bar{\gamma}_{i,2} e_{i,2}| + e_2^*} \right. \\
&\quad \left. - \|\tilde{\mathbf{H}}_{i2}\|^2 \left(-\ell_{i,2} \|\hat{\mathbf{H}}_{i2}\|^2 + \frac{|e_{i,2}|}{2} \right) \right\} + \mathbf{e}_2^T \mathbf{M}^{-1} \mathbf{K} \mathbf{e}_3 - \frac{3\mathbf{e}_2^T \mathbf{e}_2}{2} \\
&\leq \sum_{i=1}^n \left\{ \left(\frac{\beta_2 + \|\mathbf{H}_{i2}\|^2}{2} + \varepsilon_{i,2} \right) |e_{i,2}| - \frac{|\bar{\gamma}_{i,2} e_{i,2}|^2}{|\bar{\gamma}_{i,2} e_{i,2}| + e_2^*} + \ell_{i,2} \|\tilde{\mathbf{H}}_{i2}\|^2 \|\hat{\mathbf{H}}_{i2}\|^2 - \frac{\|\tilde{\mathbf{H}}_{i2}\|^2 |e_{i,2}|}{2} \right\} \\
&\quad + \mathbf{e}_2^T \mathbf{M}^{-1} \mathbf{K} \mathbf{e}_3 - \frac{3\mathbf{e}_2^T \mathbf{e}_2}{2} \\
&\leq \sum_{i=1}^n \left\{ -\zeta |e_{i,2}|^{2-\kappa} \exp(|e_{i,2}|^\kappa) + \frac{d_{i,2} (\beta_2 + \|\mathbf{H}_{i2}\|^2)}{2} + e_2^* + \frac{\bar{\varepsilon}_{i,2}^2}{2} + \frac{\ell_{i,2} \|\mathbf{H}_{i2}\|^2}{2} + \frac{d_{i,2}^2}{2\ell_{i,2}} \right\} \\
&\quad + \frac{\bar{k}^2 \mathbf{e}_3^T \mathbf{e}_3}{2} - \frac{\mathbf{e}_2^T \mathbf{e}_2}{2}
\end{aligned} \tag{4}$$

Step 3. Select $L_3 = (1/2)\mathbf{e}_3^T \mathbf{e}_3 + (1/2) \sum_{i=1}^n (\|\tilde{\mathbf{H}}_{i3}\|^4)$.

Case 1. $|e_{i,3}| \geq d_{i,3}$

$$\begin{aligned}
\dot{L}_3 &= \sum_{i=1}^n \left\{ e_{i,3} \left[- \left(\eta_{i,3} + \frac{\beta_3 + \|\hat{\mathbf{H}}_{i3}\|^2}{2} \right) \Phi(e_{i,3}) + \frac{\beta_3 + \|\mathbf{H}_{i3}\|^2}{2} + \varepsilon_{i,3} \right] - \frac{|\bar{\gamma}_{i,3}|^2 e_{i,3} \lceil e_{i,3} \rceil}{|\bar{\gamma}_{i,3} e_{i,3}| + e_3^*} \right. \\
&\quad \left. - \|\tilde{\mathbf{H}}_{i3}\|^2 \left(-\ell_{i,3} \|\hat{\mathbf{H}}_{i3}\|^2 + \frac{|e_{i,3}|}{2} \right) \right\} + \mathbf{e}_3^T \mathbf{e}_4 - \frac{(\bar{k}^2 + 2)\mathbf{e}_3^T \mathbf{e}_3}{2} \\
&\leq \sum_{i=1}^n \left\{ -\eta_{i,3} |e_{i,3}| - \zeta |e_{i,3}|^{2-\kappa} \exp(|e_{i,3}|^\kappa) + e_3^* + \frac{\bar{\varepsilon}_{i,3}^2}{2} + \frac{\ell_{i,3} \|\mathbf{H}_{i3}\|^4}{4} \right\} + \frac{\mathbf{e}_4^T \mathbf{e}_4}{2} - \frac{\bar{k}^2 \mathbf{e}_3^T \mathbf{e}_3}{2}
\end{aligned} \tag{5}$$

Case 2. If $|e_{i,3}| < d_{i,3}$, it follows that

$$\begin{aligned}
\dot{L}_3 &\leq \sum_{i=1}^n \left\{ \frac{d_{i,3}(\beta_3 + \|\mathbf{H}_{i3}\|^2)}{2} + |e_{i,3}|\varepsilon_{i,3} - \frac{|\tilde{\gamma}_{i,3}|^2 e_{i,3} \lceil e_{i,3} \rceil}{|\tilde{\gamma}_{i,3} e_{i,3}| + e_3^*} - \|\tilde{\mathbf{H}}_{i3}\|^2 \left(-\ell_{i,3} \|\hat{\mathbf{H}}_{i3}\|^2 + \frac{|e_{i,3}|}{2} \right) \right\} \\
&\quad + \mathbf{e}_3^T \mathbf{e}_4 - \frac{(\bar{k}^2 + 2) \mathbf{e}_3^T \mathbf{e}_3}{2} \\
&\leq \sum_{i=1}^n \left\{ -\zeta |e_{i,3}|^{2-\kappa} \exp(|e_{i,3}|^\kappa) + \frac{d_{i,3}(\beta_3 + \|\mathbf{H}_{i3}\|^2)}{2} + e_3^* + \frac{\bar{\varepsilon}_{i,3}^2}{2} + \frac{\ell_{i,3} \|\mathbf{H}_{i3}\|^4}{2} + \frac{d_{i,3}^2}{2\ell_{i,3}} \right\} \\
&\quad + \frac{\mathbf{e}_4^T \mathbf{e}_4}{2} - \frac{\bar{k}^2 \mathbf{e}_3^T \mathbf{e}_3}{2}
\end{aligned} \tag{6}$$

Step 4. Choose $L_4 = (1/2) \mathbf{e}_4^T \mathbf{e}_4 + (1/2) \sum_{i=1}^n (\|\tilde{\mathbf{H}}_{i4}\|^4)$.

Case 1. If $|e_{i,4}| \geq d_{i,4}$, $\dot{L}_4$ is calculated as

$$\begin{aligned}
\dot{L}_4 &= \sum_{i=1}^n \left\{ e_{i,4} \left[- \left(\eta_{i,4} + \frac{\beta_4 + \|\hat{\mathbf{H}}_{i4}\|^2}{2} \right) \Phi(e_{i,4}) + \mathbf{H}_{i4}^T \boldsymbol{\psi}_4(\mathbf{N}_4) + \varepsilon_{i,4} - \zeta \lceil e_{i,4} \rceil^{1-\kappa} \exp(|e_{i,4}|^\kappa) \right] \right. \\
&\quad \left. - \|\tilde{\mathbf{H}}_{i4}\|^2 \left(-\ell_{i,4} \|\hat{\mathbf{H}}_{i4}\|^2 + \frac{|e_{i,4}|}{2} \right) \right\} - \mathbf{e}_4^T \mathbf{e}_4 + \mathbf{e}_4^T (\mathbf{J}^{-1} \mathbf{R}(\boldsymbol{\xi}) (\mathbf{J}^{-1})^T \mathbf{N}(\boldsymbol{\varrho}) \boldsymbol{\tau}_v - \boldsymbol{\tau}_v) \\
&\leq \sum_{i=1}^n \left\{ - \left(\eta_{i,4} - \frac{\|\tilde{\mathbf{H}}_{i4}\|^2}{2} \right) |e_{i,4}| - \zeta |e_{i,4}|^{2-\kappa} \exp(|e_{i,4}|^\kappa) + \ell_{i,4} \|\tilde{\mathbf{H}}_{i4}\|^4 + \frac{\ell_{i,4} \|\mathbf{H}_{i4}\|^2}{4} - \ell_{i,4} \|\tilde{\mathbf{H}}_{i4}\|^4 \right. \\
&\quad \left. - \frac{\|\tilde{\mathbf{H}}_{i4}\|^2 |e_{i,4}|}{2} \right\} - \frac{\mathbf{e}_4^T \mathbf{e}_4}{2} + \frac{\boldsymbol{\varepsilon}_4^T \boldsymbol{\varepsilon}_4}{2} + \mathbf{e}_4^T (\mathbf{F}(\boldsymbol{\xi}) \mathbf{N}(\boldsymbol{\varrho}) \boldsymbol{\tau}_v - \boldsymbol{\tau}_v) \\
&\leq \sum_{i=1}^n \left\{ -\eta_{i,4} |e_{i,4}| - \zeta |e_{i,4}|^{2-\kappa} \exp(|e_{i,4}|^\kappa) + \frac{\bar{\varepsilon}_{i,4}^2}{2} + \frac{\ell_{i,4} \|\mathbf{H}_{i4}\|^4}{4} + \frac{1}{g_i} (-F_i(t) N(\varrho_i) + 1) \dot{\varrho}_i \right\} - \frac{\mathbf{e}_4^T \mathbf{e}_4}{2}
\end{aligned} \tag{7}$$

Case 2. When $|e_{i,4}| < d_{i,4}$, $\dot{L}_4$ is given by

$$\begin{aligned}
\dot{L}_4 &= \sum_{i=1}^n \left\{ e_{i,4} \left[- \left(\eta_{i,4} + \frac{\beta_4 + \|\hat{\mathbf{H}}_{i4}\|^2}{2} \right) \Phi(e_{i,4}) + \mathbf{H}_{i4}^T \boldsymbol{\psi}_4(\mathbf{N}_4) + \varepsilon_{i,4} - \zeta \lceil e_{i,4} \rceil^{1-\kappa} \exp(|e_{i,4}|^\kappa) \right] \right. \\
&\quad \left. - \|\tilde{\mathbf{H}}_{i4}\|^2 \left(-\ell_{i,4} \|\hat{\mathbf{H}}_{i4}\|^2 + \frac{|e_{i,4}|}{2} \right) \right\} - \mathbf{e}_4^T \mathbf{e}_4 + \mathbf{e}_4^T (\mathbf{J}^{-1} \mathbf{R}(\boldsymbol{\xi}) (\mathbf{J}^{-1})^T \mathbf{N}(\boldsymbol{\varrho}) \boldsymbol{\tau}_v - \boldsymbol{\tau}_v) \\
&\leq \sum_{i=1}^n \left\{ \frac{d_{i,4}(\beta_4 + \|\mathbf{H}_{i4}\|^2)}{2} - \zeta |e_{i,4}|^{2-\kappa} \exp(|e_{i,4}|^\kappa) + \ell_{i,4} \|\tilde{\mathbf{H}}_{i4}\|^2 \|\hat{\mathbf{H}}_{i4}\|^2 - \frac{\|\tilde{\mathbf{H}}_{i4}\|^2 |e_{i,4}|}{2} \right\} \\
&\quad - \frac{\mathbf{e}_4^T \mathbf{e}_4}{2} + \frac{\boldsymbol{\varepsilon}_4^T \boldsymbol{\varepsilon}_4}{2} + \mathbf{e}_4^T (\mathbf{F}(\boldsymbol{\xi}) \mathbf{N}(\boldsymbol{\varrho}) \boldsymbol{\tau}_v - \boldsymbol{\tau}_v) \\
&\leq \sum_{i=1}^n \left\{ -\zeta |e_{i,4}|^{2-\kappa} \exp(|e_{i,4}|^\kappa) + \frac{d_{i,4}(\beta_4 + \|\mathbf{H}_{i4}\|^2)}{2} + \frac{\bar{\varepsilon}_{i,4}^2}{2} + \frac{\ell_{i,4} \|\mathbf{H}_{i4}\|^4}{2} + \frac{d_{i,4}^2}{2\ell_{i,4}} \right. \\
&\quad \left. + \frac{1}{g_i} (-F_i(t) N(\varrho_i) + 1) \dot{\varrho}_i \right\} - \frac{\mathbf{e}_4^T \mathbf{e}_4}{2}
\end{aligned} \tag{8}$$

$F_i(t)$ is bounded by some unknown set.